

Step 3 Results Note

Pretrained Masked-Language-Model Reconstruction: Clean vs. OCR Context

Field	Detail
Problem ID	ML-T2-022
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Stage	Stage 2, Step 3 — MLM-based text reconstruction (Approach B, pretrained baseline)

1. Purpose

Establish a pretrained (no fine-tuning) masked-language-model baseline for reconstructing single missing words, before investing time in fine-tuning on Kaggle. Per the Stage 2 build philosophy, the cheap path (off-the-shelf RoBERTa) is tested before the expensive path (fine-tuning) is justified.

2. Method

- Word-level ground truth positions taken from Tesseract on the clean image (per the correction that a fully torn word may leave no OCR box on the damaged image itself).
- Each of the 19 words in the test sentence was masked one at a time, using a validated Step 2 word-targeted mask (single-word precision confirmed: 17/17 and 16/16 untouched words preserved in prior validation).
- Masking is done by word position/index, not by text match, after an early bug caused every occurrence of a repeated word ("everything", appearing twice) to be masked simultaneously, corrupting the pipeline's output structure.
- Two context conditions tested: (a) clean-image context, every surrounding word is ground-truth correct; (b) OCR-derived context, surrounding words come from Tesseract's reading of the actual torn image, which may itself contain errors near the tear.
- Pretrained roberta-base fill-mask pipeline used, top-5 predictions requested per word, no fine-tuning applied at this stage.

3. Results

Context condition	Top-1 accuracy	Top-5 accuracy
Clean-image context (ground truth)	52.63%	78.95%
OCR-derived context (damaged image)	47.37%	78.95%

18 of 19 words produced identical results across both conditions, since the validated single-word masks rarely caused OCR errors beyond the targeted word itself. One word ("believe") diverged: the tear on "believe" caused a knock-on OCR misread of the adjacent word "to" (read as "tc"), which changed the surrounding context enough that the model's top prediction shifted from the correct word ("believe", low confidence 0.173) to an incorrect one ("doubt", 0.377).

Across both conditions, function words and highly grammatically-constrained words ("To", "everything", "with", "the", "of") were predicted correctly with high confidence (0.55–0.99). Content words carrying the sentence's specific meaning ("convenient", "solutions;", "dispense", "necessity", "reflection.") were consistently missed at top-1, and notably scored lower confidence on their top prediction (0.13–0.30) than the correctly-predicted words.

4. Interpretation

Two findings, both relevant to the project's central research question:

- Reconstruction accuracy is uneven by word type: pretrained RoBERTa reliably recovers function words but struggles with content-bearing words, exactly the words most likely to matter in a real archival document (names, dates, key terms).
- Model confidence correlates with correctness in this small sample: low-confidence predictions were consistently the wrong ones. This is an early, weak signal (19 words, one sentence) but directionally supports the feasibility of a confidence-threshold-based uncertainty layer (Approach C) rather than requiring a separately trained calibration model.

A caveat on the clean-vs-OCR-context comparison: this test used isolated single-word tears on an otherwise undegraded image, which Step 2 validation showed rarely bleeds into neighboring OCR reads. The gap between clean and OCR context is expected to widen once torn regions are combined with Step 1's faded/noisy degradation, where OCR is already unreliable near the tear before the physical gap is introduced. This has not yet been tested and is noted as a next step rather than a current finding.

5. Decision: Fine-Tuning

Given the pretrained baseline's weak performance specifically on content words (the category most relevant to archival use), fine-tuning on Kaggle (GPU) is judged worthwhile rather than optional. This baseline result is the evidence justifying that additional cost, per the build philosophy of not paying for the expensive path before the cheap path is shown to be insufficient.

6. Next Steps

- Repeat this test combining torn regions with Step 1's faded/noisy degradation, to check whether the clean-vs-OCR-context gap widens as expected.
- Expand the test set beyond one sentence before treating the confidence-correctness correlation as a reliable finding rather than a preliminary signal.
- Move to Kaggle for RoBERTa fine-tuning on masked-span prediction, using the mask-generation and damage-classification pipeline built in Steps 2–3 to produce training data.